Big Data Analytics & Text Mining

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Problem and Dataset Description

- Clickstream Data for Online Shopping
  - UCI Machine Learning Repository
  - Single .csv file containing 165k observations of 14 features

- The dataset contains information on clickstream from online store offering clothing for pregnant women. Data are from five months of 2008 and include, among others, product category, location of the photo on the page, country of origin of the IP address and product price in US dollars.

Problem and Dataset Description

- Task
  - Clustering
  - Classification
  - Regression

- Notebook Execution
  - The notebook was entirely created and executed via Google Colab, using the CPU resource that is available on the platform. The main characteristics for this VM made available by Google are: 12GiB RAM available, Intel(R) Xeon(R) CPU @ 2.00GHz processor

Dataset - Variables

Variables involved in the analysis

Variable	Type	Variable	Type
Year	Numerical	Page 2	**Categorical**
Month	Numerical	Colour	Numerical
Day	Numerical	Location	Numerical
Order	Numerical	Model Photography	Numerical
Country	Numerical	Price	Numerical
Session ID	Numerical	Price 2	Numerical
Page 1	Numerical	Page	Numerical

The only categorical variable present in the dataset (page 2) is encoded to be included in the analysis.

CLUSTERING TASK

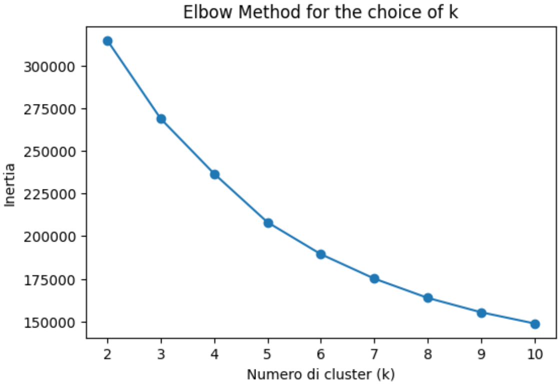
Key concept: clickstream data grouped by session ID

The main variables used for clustering are the following:

Variable	**Type**	**Variable**	**Type**
session_length	Num	category_switch_count	Num
num_products	Num	mean_price	Num
num_categories	Num	std_price	Num
dominant_cat_ratio	Num	price_trend	Num

Elbow Method

Clustering Task

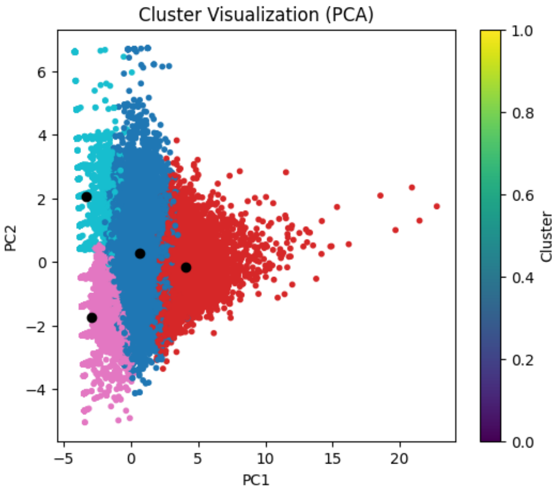


The value chosen for k is 4.

The choice of k=5 would have been adequate too.

Results and Visualization

Clustering Task



Cluster 1

colour blue in the PCA visualization

- medium sessions on the website (session_length = 5.47 on average, num_products close to 5 on average), visiting about 2 categories of products (num_categories = 1.86 on average) with a single dominant category (dominant_cat_ratio = 0.75).
- mean_price has an average of 45 USD and a standard deviation of 12. Price trend is on average close to -0.15.

Label: curious explorers that visit clothes with a progressive price downgrade, probably searching for occasions and discounts for a particular category of products.

Cluster 2

colour red in the PCA visualization

- longer sessions (session_length = 18.8 on average, num_products close to 16.5 on average), about 3.17 categories on average and particularly active users (category_switch_count = 3.13 on average, clicks_per_page close to 6.48 on average).
- average price similar to the one of the first group (mean_price = 43 USD on average) and the price trend is also negative on average, close to -0.32.

Label: dynamic users that are not looking for something in particular but instead visit the shop in search for occasions that they recognize by comparing prices between different products.

Cluster 3

colour pink in the PCA visualization

- minimal sessions (session_length = 1.8, num_products = 1.67 on average), in most cases just one or two products in the same category (num_categories = 1.1, dominant_cat_ratio = 0.95 on average).
- mean_price a little lower with respect to the first 2 clusters and very stable (mean_price = 36, std_price = 1.7, price_trend = -0.08 on average).

Label: focused buyers that are already oriented to the products they want to view and probably have a rapid decisional behaviour too.

Cluster 4

colour light blue in the PCA visualization

- minimal sessions (session_length = 1.6, num_products = 1.47 on average) and also visit just one category without switching (num_categories = 1.05, dominant_cat_ratio = 0.98, category_switch_count = 0.05 on average).
- main difference with the third cluster stands in the price, both in levels and trend (mean_price = 56, price_trend = +0.43 on average).

Label: premium buyers looking for high quality products, with a similar decisional behaviour to the third cluster but higher budget for shopping.

CLASSIFICATION TASK

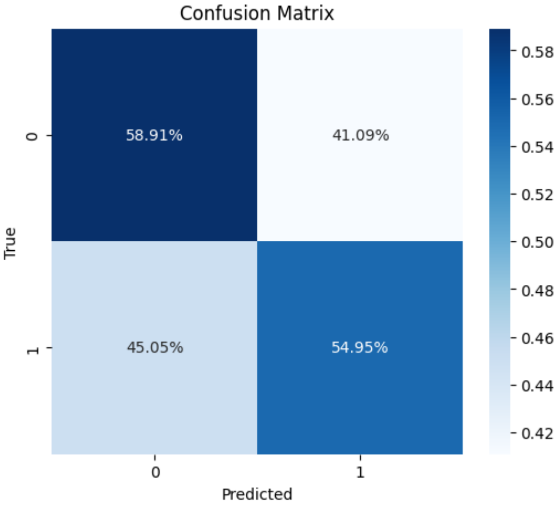
Binary Classification Task
Target Variable

- Target Variable: price 2
- Supervised Binary Classification

Models - Logistic Regression

Linear Classifiers

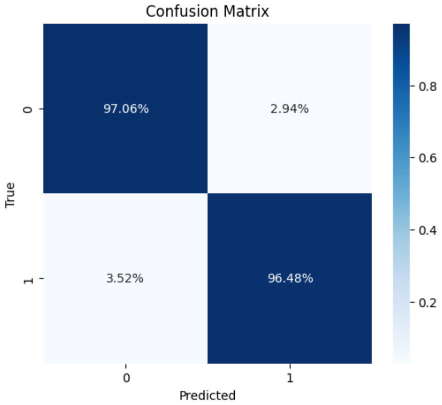
- Accuracy: 56.98%



Models - Random Forest

Non-linear Classifiers

- Accuracy: 96.78%



Multi-Class Classification Task

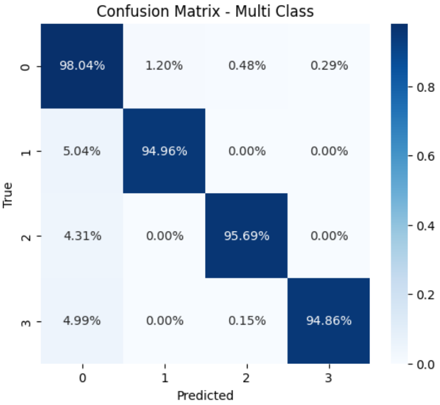
Target Variable

- Target Variable: price
- Supervised Multi-Class Classification

Models - RandomForest

Non-linear Classifiers

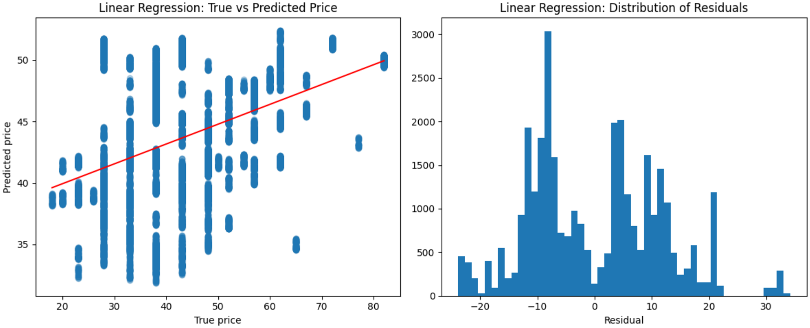
- Accuracy: 96.44%



REGRESSION TASK

Linear Regression
Regression Task

Target column = price

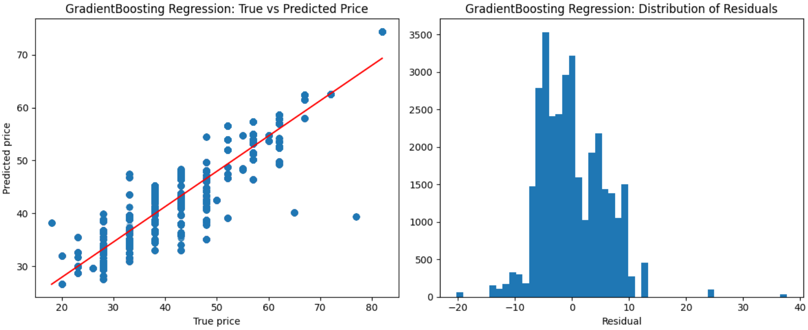


MAE = 9.86 RMSE = 11.51 R^2 = 0.164

Gradient Boosting Regression

Regression Task

Target column = price



MAE = 4.46 RMSE = 5.63 R^2 = 0.800

Conclusion: discussion of results

- Clustering Task: effective clusters even if Silhouette Score is lower than expected and PCA visualization shows overlapping. (Common when working with clickstream data).
- Classification Task: great difference in performance between linear classifiers and tree classifiers, due to ability to capture non-linear correlation between data.
- Regression Task: similarities with classification task even in a continuous framework.

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